

PRAM■■■A

Project Architecture

Evidence-gated enterprise knowledge multi-agent · Team KMIT · Ruthvik Goud (Backend) · Monisha Sarai (Frontend) · Rohith (DevOps)

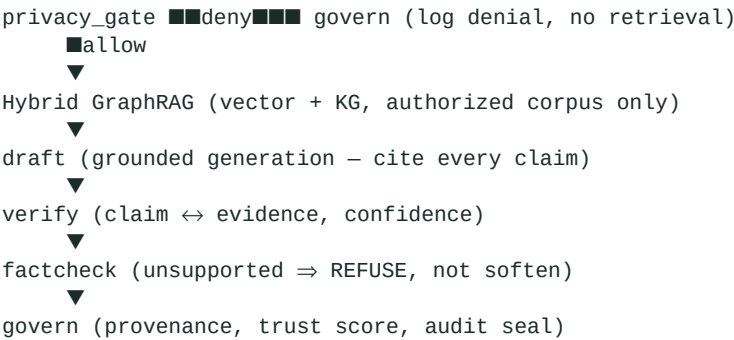
Truth is not assumed. It is proven. · Deny / refuse = PASS

1. Mission & design invariants

PRAM■■■A is a trust-first enterprise knowledge agent. It authorizes **before** retrieval, grounds every claim in authorized evidence, and treats policy-correct denial and hallucination refusal as **successful** outcomes — not soft failures.

- **No retrieve-then-filter** — privacy gate must resolve before any corpus call.
- **Named single-responsibility agents only** — no god-agent.
- **Deterministic-first** for gate / verify / factcheck; LLMs are additive polish.
- **Judge never mutates the agent** — Helix Evaluate is judge-only.
- **Govern always runs last** — on allow and deny paths.

2. Fixed trust pipeline



Short-circuit: when privacy_gate denies, retriever / draft / verify / factcheck are **skipped**; govern still seals the audit.

3. Named agents

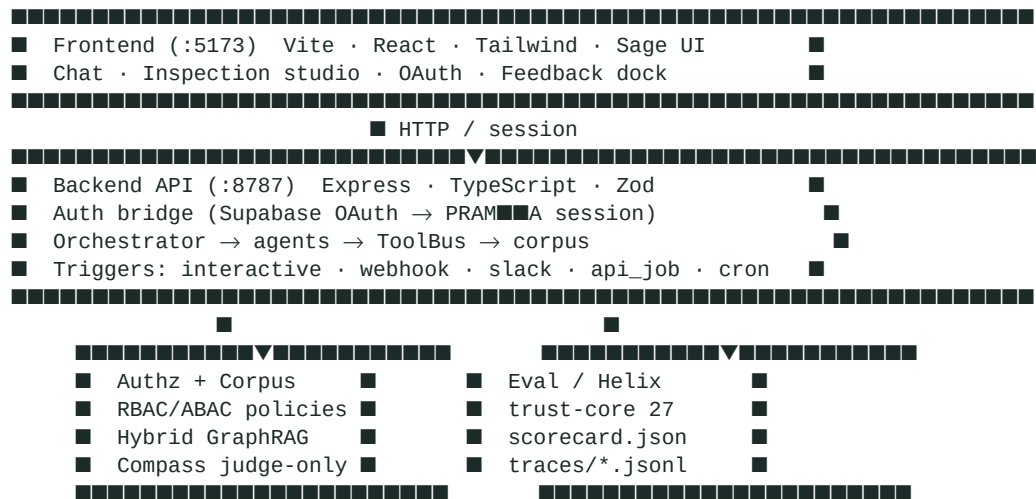
Agent	Role	Responsibility
privacy_gate	gate	RBAC + ABAC + adversarial resistance; deny before retrieve
retriever	retrieve	Ticket-scoped Hybrid GraphRAG (vector + KG expand)
draft	generate	Draft only from authorized hits; cite source docs
verify	verify	Claim extraction + evidence binding + confidence
factcheck	validate	Hallucination scan; unsupported ⇒ REFUSE
govern	audit	Provenance, trust score, full hop audit (always last)

4. Tooling bus (audited jobs)

Tools are first-class audited jobs on a bus — every call is observable in hops / audit.

- policy.check · corpus.search · graph.expand
- claims.extract · evidence.bind · hallucination.scan
- audit.seal · notify.compliance · compass.verify (judge-only)

5. System architecture (layers)



6. Repository map

Path	Purpose
submissions/pramana/backend/	Multi-agent pipeline, auth, API, EDD runners
submissions/pramana/frontend/	Sage landing + chat + inspection UI
submissions/pramana/eval/	Dataset, runner, scorecard
submissions/pramana/traces/	Helix-consumable JSONL traces
submissions/pramana/transcripts/	Native Helix / evolve session packs
submissions/pramana/helix-extension/	*govern skill (trust vs quality)
submissions/pramana/files/	Judge pack (this PDF + audit + guides)
agentspec.yaml	Portable MultiAgent definition
.mutagent/ · .claude/ · .agents/	Helix / Mutagent install + mirrored eval

7. Auth & product surface

- Supabase OAuth (Google / GitHub) → POST /auth/supabase session bridge.
- Clearance roles: employee · analyst · manager · compliance (and dept / level ABAC).
- Chat workspace with live inspection: hops, citations, trust score.
- Optional LLM polish (OpenRouter / Anthropic) only **after** gate allow + grounded draft.
- Feedback dock → mutagent feedback send.

8. Helix / Mutagent lifecycle

*mutagent → *evaluate → *govern → SHIP
 ■ *diagnose → *optimize (approval-gated)

Self-evolve proof: 18/27 FAIL → diagnose → apply → 27/27 SHIP
(npm run edd:evolve)

Extension ***govern / pramana-govern** sits between Evaluate and Diagnose. It classifies TRUST-DECISION vs OUTPUT-QUALITY failures. Judge-never-fix holds.

9. Five binary eval criteria (architecture contract)

id	Pass means
authz_deny_before_retrieve	Denied queries never call the retriever
citation_grounding	Every claim maps to ≥ 1 authorized evidence id
hallucination_refuse	Unsupported $\Rightarrow$ REFUSE, not hedge
refusal_is_success	Expected refuse/deny fixtures score PASS
audit_completeness	Every hop logged (agent, decision, evidence refs)

Deliverable root: submissions/pramana/ · Spec: agentspec.yaml · Stack: Node/Express/TS + React/Vite